

Oppgave 1

a)

$$R_B = 220k\Omega$$

$$R_C = 470\Omega$$

$$V_{cc} = 5V$$

$$\beta = 330$$

$$V_B = V_{BE} = 0.7V$$

$$I_B = \frac{V_{cc} - V_{BE}}{R_B} = \frac{4.3}{220k} = 195\mu A$$

$$I_C = \beta \cdot I_B = 195\mu \cdot 330 = 6.45mA$$

$$V_C = V_{cc} - R_C \cdot i_C = 5 - 470 \cdot 6.45\mu = 1.97V$$

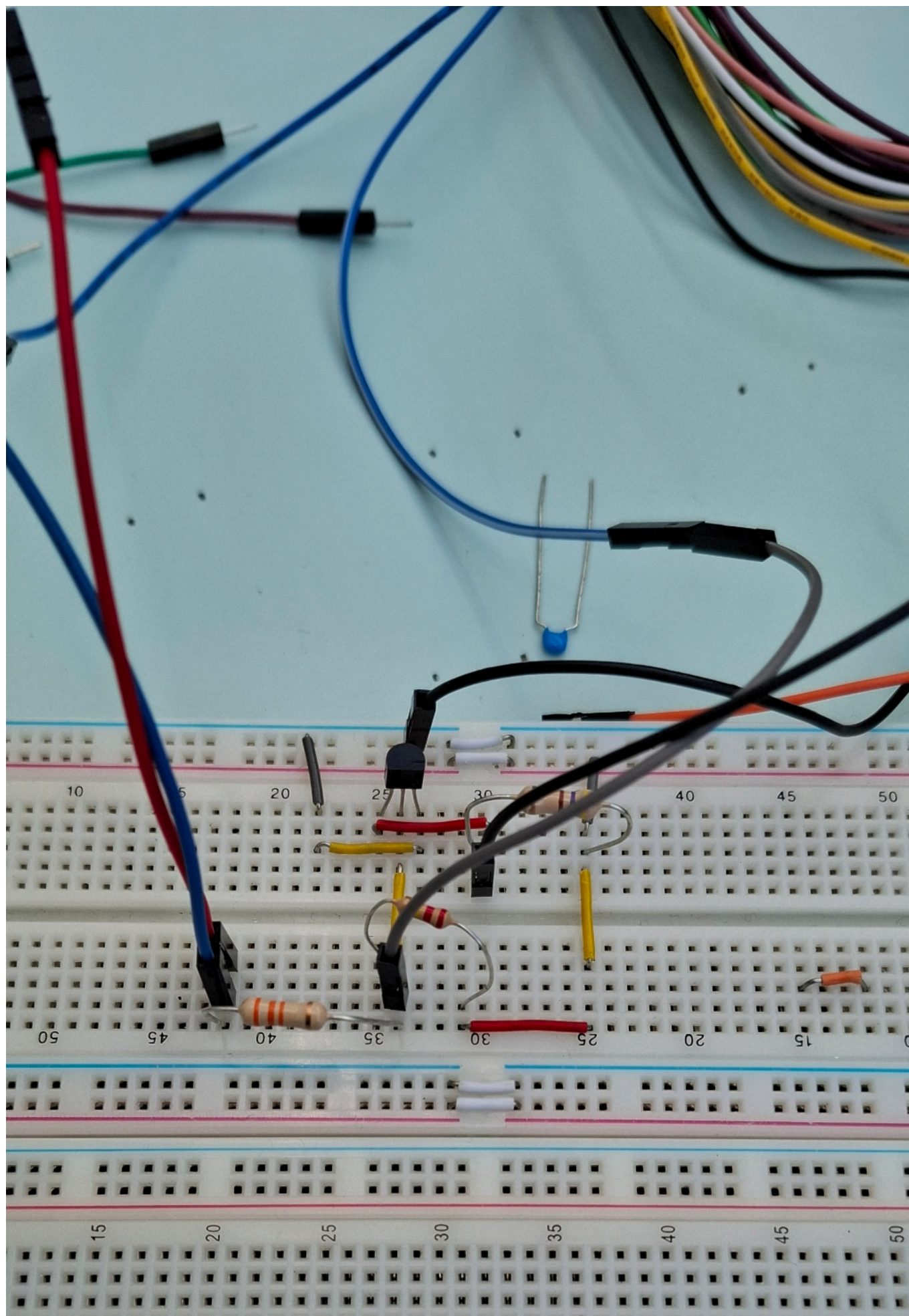
$$R_o = R_C = 470\Omega$$

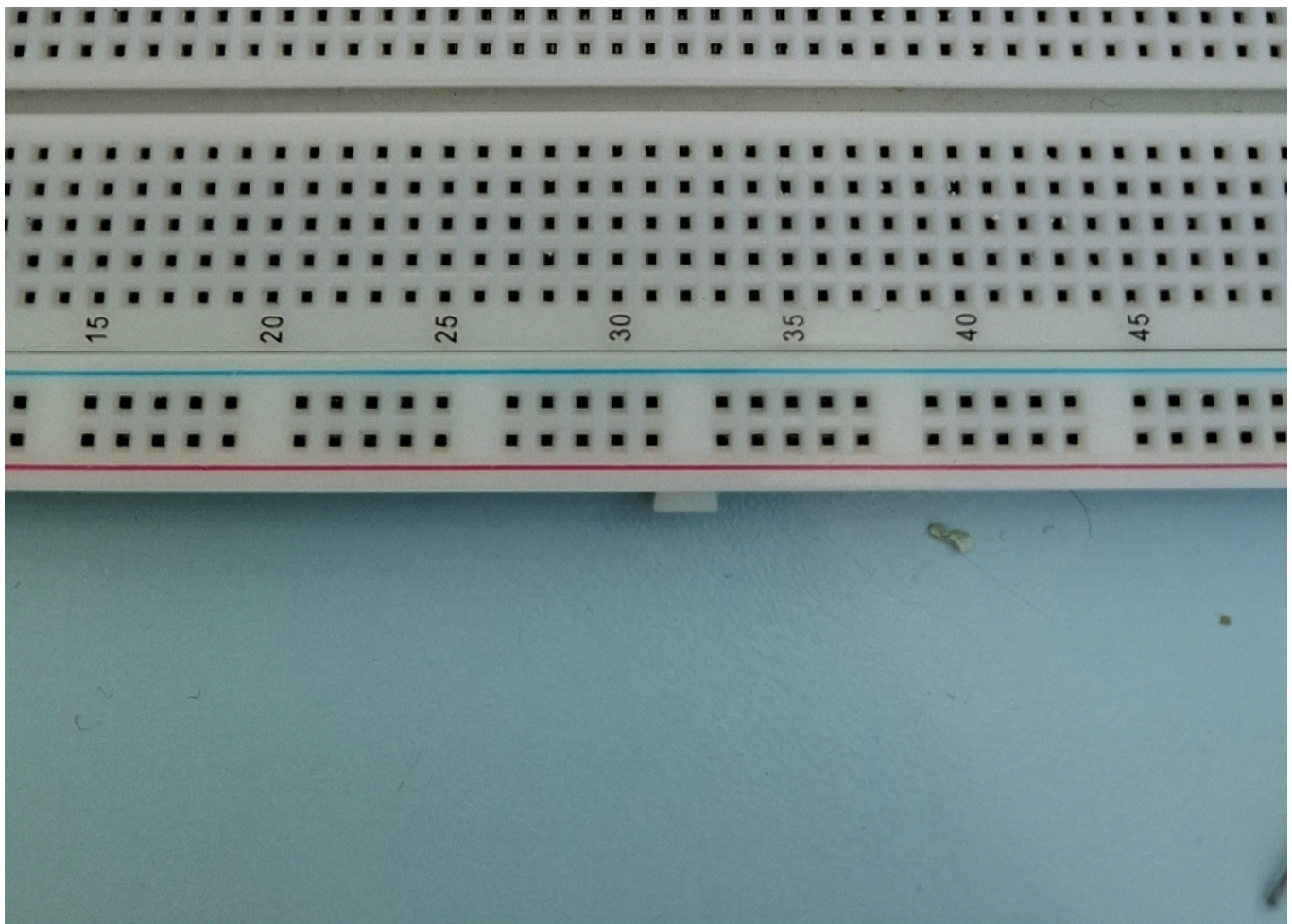
$$R_i = R_B || r_\pi \approx 4.9k \text{ med en } r_\pi = 5k\Omega$$

$$A = \beta \frac{R_C}{r_\pi} = -330 \cdot \frac{470}{5k} = 31.08$$

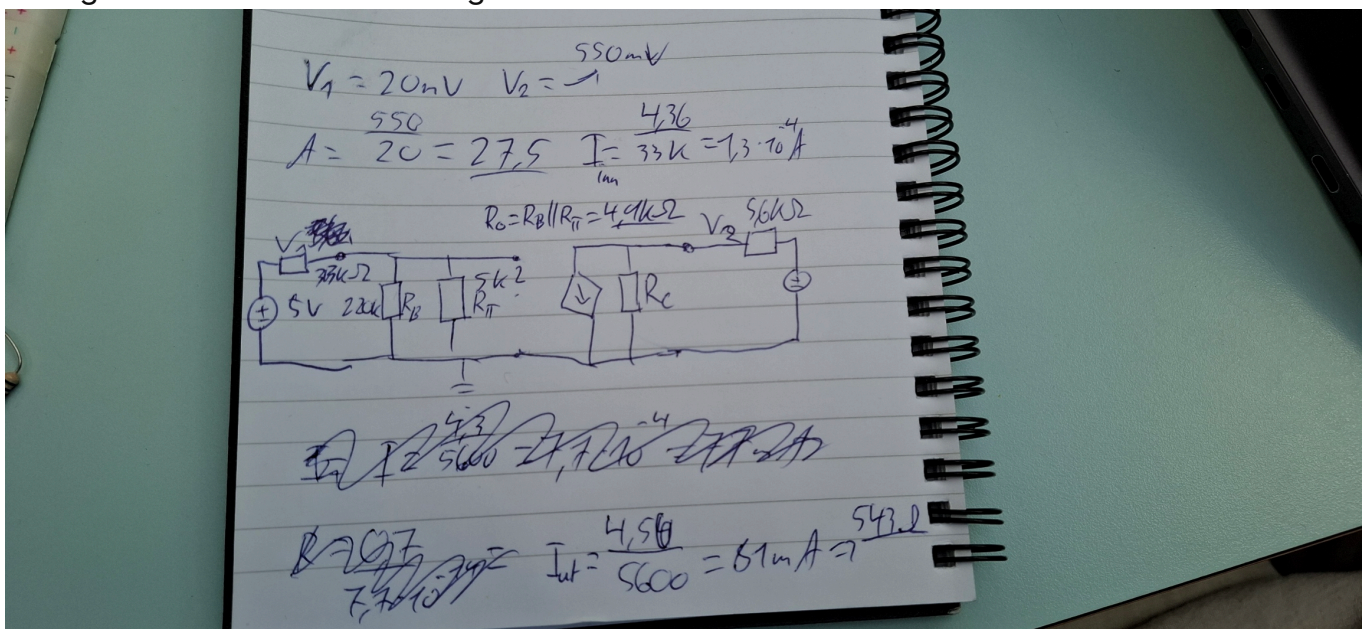
b)

Oppkobling:





Veldig rotete notater for kobling:

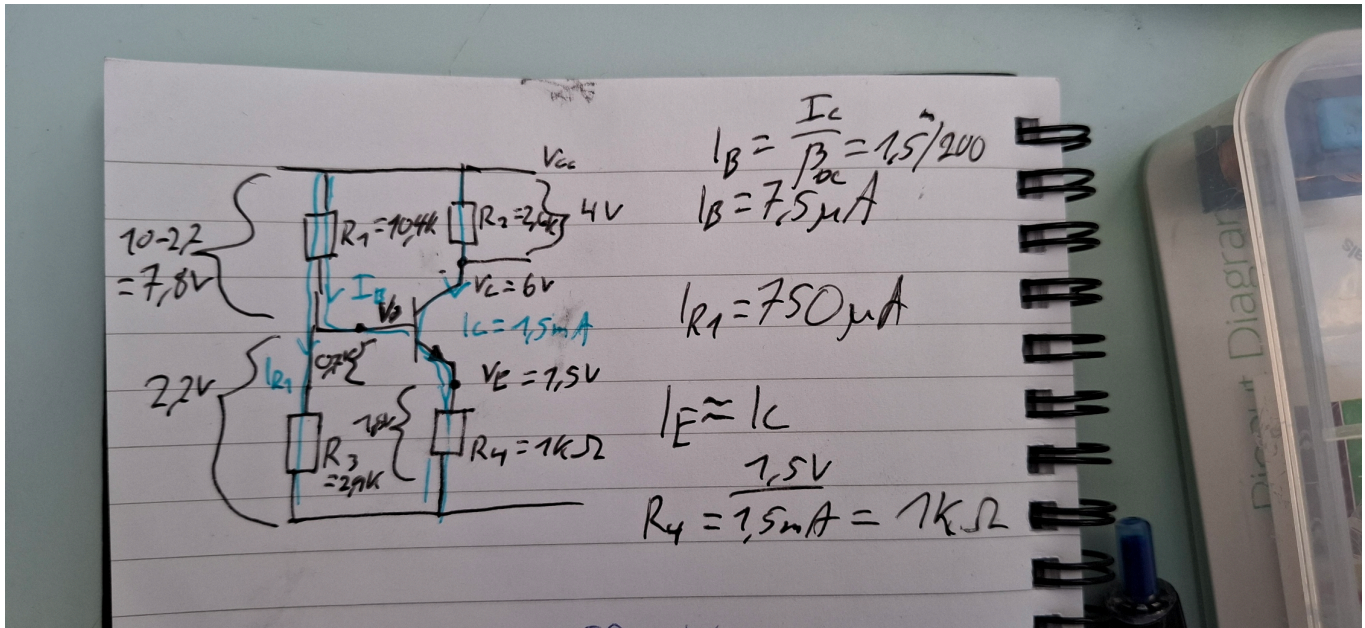


Koblet en $33\text{k}\Omega$ motstand og 5V på inngangen og målte spenningen over den til 4.36V. Regner så ut $R_O = R : B || R_{\pi} \approx 4.9\text{k}\Omega$.

Samme på utgangen med en

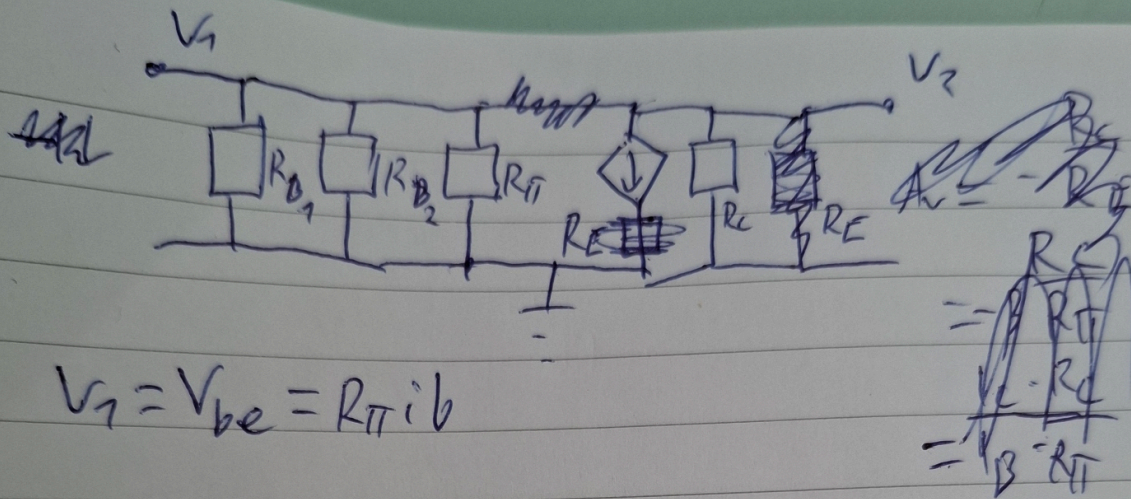
Oppgave 2)

a)



$$\begin{aligned} R_1 &= 10k\Omega \\ R_2 &= 2.7k\Omega \\ R_3 &= 2.7k\Omega \\ R_4 &= 1k\Omega \end{aligned}$$

b)



$$V_1 = V_{be} = R_{\pi} i_b$$

$$V_2 = -R_{CL} \cdot \beta i_b$$

$$R_{CL} = \beta i_b$$

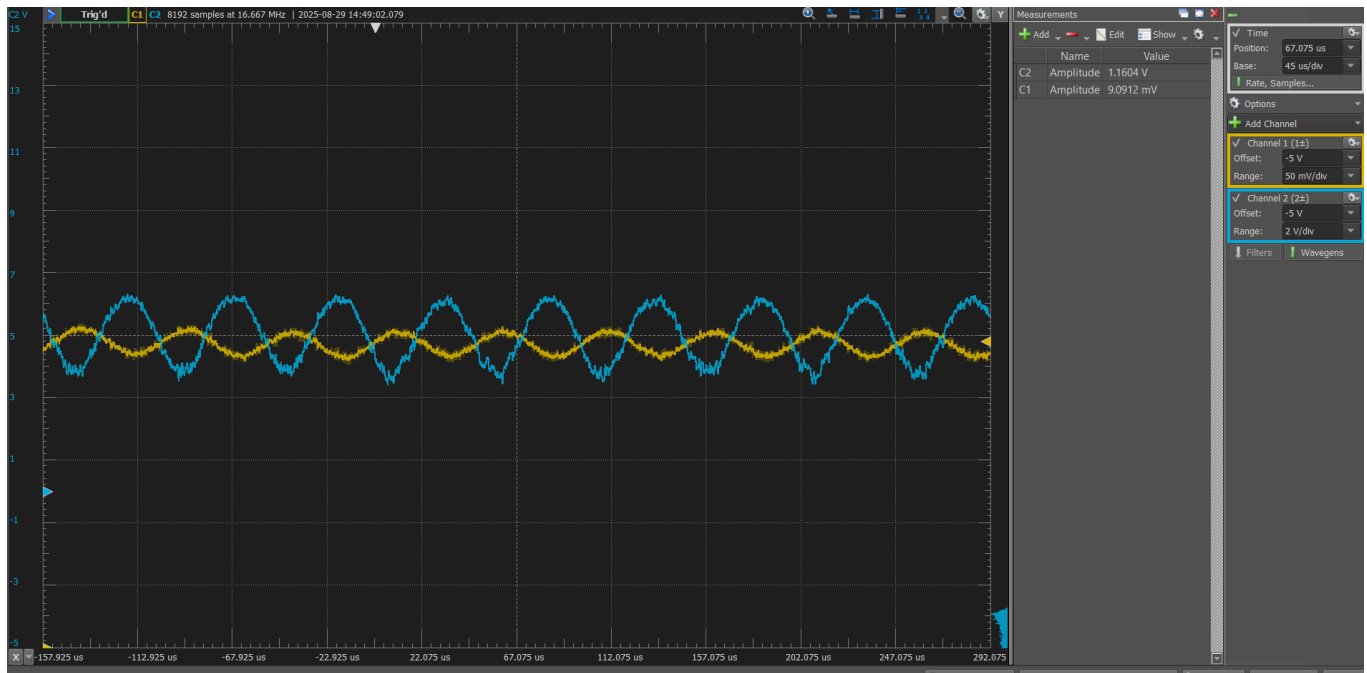
$$A_v = - \frac{R_{CL}}{R_{\pi} i_b}$$

$$\beta = 330 \quad R_{\pi} \approx 5k \quad (3,2 - 8,5)$$

$$\text{Without load: } R_{CL} = R_C = 2,7k$$

$$A_v = 1782 \quad [105, 278]$$

c)



$$A_v \approx \frac{1.16}{0.009} = 129$$

Ja verdien er innenfor området på transistoren, og måtte gjøre en del avrundinger på motstandene så usikkerheten er ganske høy.

Oppkobling:

